

- 1 (a) The European eel, *Anguilla anguilla*, is a fish. The sizes of eel populations tend to remain relatively stable despite eels producing large numbers of offspring.

Suggest **two** reasons why the population sizes of eels tend to remain relatively stable.

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- (b) Explain what is meant by the general theory of evolution.

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- (c) The generation time of a species is the mean (average) time from one generation (parents) to the next generation (offspring). For example, the generation time of humans is about 25 years.

Fig. 1.1 shows a graph of the relationship between the rate of evolution and the generation time for a wide range of different species.

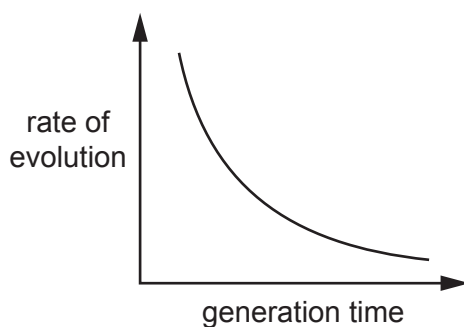


Fig. 1.1

Describe **and** explain the relationship shown in Fig. 1.1.

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- (d) The tuatara, *Sphenodon punctatus*, is a reptile that is native to New Zealand. It is found nowhere else in the wild.

Fig. 1.2 shows a tuatara.



Fig. 1.2

Tuatara have a slow growth rate and can live for over one hundred years. Fossil evidence shows that there has been little morphological change in the tuatara over the last 200 million years. This is a much lower rate of evolution than would be expected from the generation time of this species.

Suggest **and** explain why the tuatara has remained largely unchanged over the last 200 million years.

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[Total: 10]